Topological DNL Magnons in Layered Honeycomb Antiferromagnets and the Role of Interlayer Spin Canting

Samriddha Ganguly, Shayan Garai, Nirmal Ganguli
Department of Physics, Indian Institute of Science Education and Research (IISER) Bhopal

Objectives

- Derive the magnon Hamiltonian for a layered honeycomb Néel AFM.
- Show how combined time-reversal and inversion ($\mathcal{PT}$) symmetry protect Dirac nodal line (DNL) crossings.
- Characterize the DNL by a **quantized Berry phase** and vanishing Berry curvature.
- Introduce an **interlayer spin-canting interaction** and show how it gaps the DNL by breaking $\mathcal{PT}$.

Magnons and Dirac Nodal Lines

- Magnons: quantized spin waves, charge-neutral carriers of spin and heat.
- Topological magnon bands ⇒ robust edge modes and Hall responses.

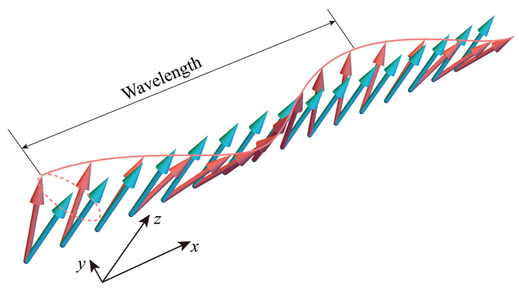

Figure 1: Spin-wave precession: classical picture of a magnon mode.

Layered Honeycomb AFM: Spin Model

Collinear Néel order on A/B sublattices:

$$\boldsymbol{S}_A = +S\hat{z}, \qquad \boldsymbol{S}_B = -S\hat{z}.$$

Minimal model (no canting):

$$\mathcal{H}_0 = J\sum_{\langle ij\rangle,l} \boldsymbol{S}_{i,l}\cdot\boldsymbol{S}_{j,l} + J_c\sum_{\langle ll'\rangle,i} \boldsymbol{S}_{i,l}\cdot\boldsymbol{S}_{i,l'} + \sum_{\langle\langle ij\rangle\rangle,l} \boldsymbol{D}_{ij}\cdot\left(\boldsymbol{S}_{i,l}\times\boldsymbol{S}_{j,l}\right).$$

- $J > 0$: intralayer AFM exchange.
- $J_c > 0$: interlayer AFM coupling.
- $\boldsymbol{D}_{ij} = D\,\nu_{ij}\,\hat{z}$: NNN DMI.

Why DNL Magnons in AFMs?

- Topological bosons without charge ⇒ **ultra-low Joule heating.**
- DNLs provide momentum-selective channels for magnon transport.
- Tunable via J_c, D and D_c ⇒ promising for reconfigurable magnonic devices.
- Antiferromagnets naturally support **THz-frequency magnon dynamics**, making them ideal for ultrafast spin transport.

Spin-Wave Theory

Linear Holstein–Primakoff expansion:

$$S^z_{i,A} = S - a_i^\dagger a_i, \quad S^z_{j,B} = -S + b_j^\dagger b_j,$$
$$S^+_{i,A} \approx \sqrt{2S}\,a_i, \quad S^+_{j,B} \approx \sqrt{2S}\,b_j^\dagger.$$

Quadratic Hamiltonian in momentum space:

$$\mathcal{H}_0 = \frac{1}{2}\sum_{\boldsymbol{k}} \Psi_{\boldsymbol{k}}^\dagger H(\boldsymbol{k})\Psi_{\boldsymbol{k}},$$

$$\Psi_{\boldsymbol{k}}^\dagger = \left(a_{\boldsymbol{k}A}^\dagger, a_{\boldsymbol{k}B}^\dagger, a_{-\boldsymbol{k}A}, a_{-\boldsymbol{k}B}\right).$$

Bosonic Bogoliubov diagonalization:

$T_{\boldsymbol{k}}^\dagger \sigma_z H(\boldsymbol{k}) T_{\boldsymbol{k}} = \sigma_z \operatorname{diag}\left(E_+(\boldsymbol{k}), E_-(\boldsymbol{k}), E_+(\boldsymbol{k}), E_-(\boldsymbol{k})\right)$, giving two positive magnon branches $E_\pm(\boldsymbol{k})$.

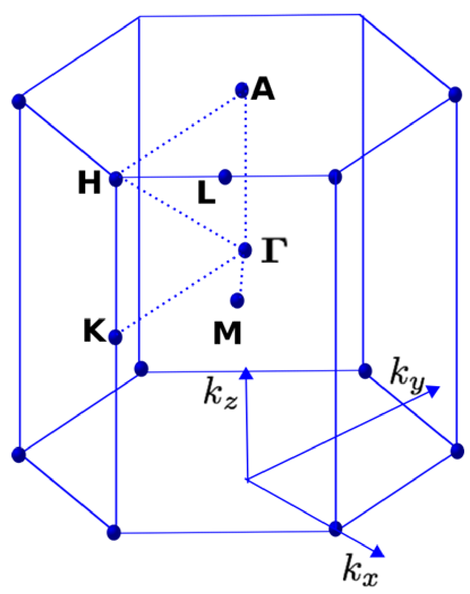

Figure 2: Hexagonal Brillouin zone and high-symmetry path M–Γ–K–H–Γ–A.

Dirac Nodal Line and PT Symmetry

- **Time reversal** $\mathcal{T}$ flips spins and $\boldsymbol{k}$; inversion $\mathcal{P}$ exchanges A/B.
- For the collinear Néel state, $\mathcal{PT}$ **leaves the ground state and** $\mathcal{H}_0$ **invariant.**

$\mathcal{PT}$ forces the Berry curvature and thus magnon thermal hall response to vanish:

$$\boxed{\Omega_n(\boldsymbol{k}) = 0 \;\Rightarrow\; \kappa_{xy} = 0.}$$

The **DNL is topological** via a loop Berry phase:

$$\boxed{\gamma_C = \oint_C \mathbf{A}_n(\mathbf{k})\cdot d\mathbf{k} = \pi}$$

for any closed path C encircling the nodal line.

Experimental Pathways

Candidate materials

- Honeycomb iridates and Ru-based compounds with strong SOC.
- Layered van der Waals AFMs such as $CrCl_3$, $NiPS_3$.

Tuning knobs

- Magnetic field → additional canting and gap control.
- Pressure/strain → modifies J_c and interlayer geometry.
- Interface engineering in heterostructures to enhance D and D_c.

Magnon Bands and Effect of Interlayer Spin Canting

DNL phase ($\mathcal{PT}$-symmetric):

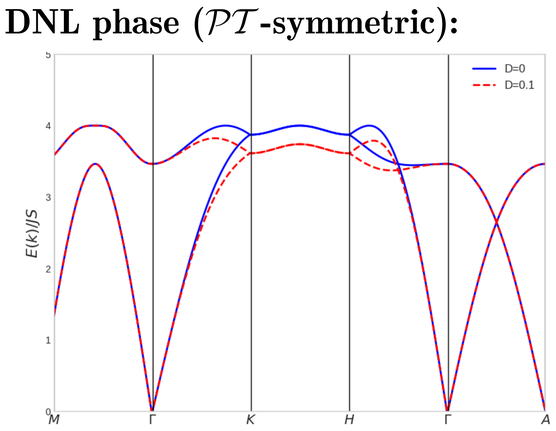

Figure 3: Magnon bands for $J_c/J = 0.5$. Blue: $D = 0$. Red dashed: finite intralayer DMI. DMI shifts bands but preserves gapless DNL along K–H due to $\mathcal{PT}$ symmetry.

Interlayer spin canting:

Add

$$\mathcal{H}_{D_c} = \sum_{\langle i,j\rangle,\langle l,l'\rangle} \boldsymbol{D}_c\cdot\left(\boldsymbol{S}_{i,l}\times\boldsymbol{S}_{j,l'}\right),$$
$$\boldsymbol{D}_c = D_c\hat{z}.$$

Physical Meaning: Physically, $\mathcal{H}_{D_c}$ arises from canting of spins in adjacent layers due to DMI and interlayer exchange. This can happen under intrinsic anisotropies

- Odd under inversion, even under $\mathcal{T}$ ⇒ breaks $\mathcal{PT}$.
- Opens a finite gap at the former nodal line.
- Berry curvature becomes nonzero ⇒ magnon Hall responses.

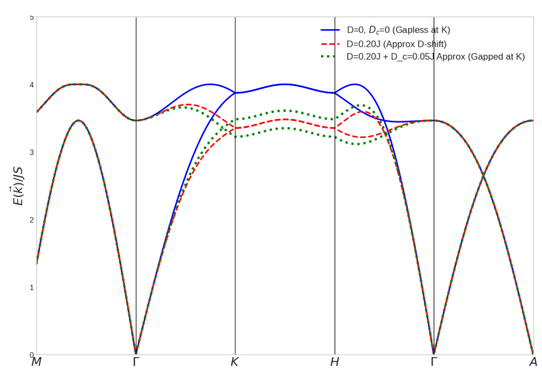

Figure 4: Blue: $D = 0, D_c = 0$ (DNL). Red dashed: D only. Green dotted: with D_c where the nodal line is fully gapped.

Why Care about Gaps? A gap allows **observable thermal Hall effects in magnon transport.** This is crucial for potential applications in magnonic devices, where **control over magnon propagation** is desired.

Key Takeaways

- Layered honeycomb AFMs can host **$\mathcal{PT}$-protected Dirac nodal line** magnons with vanishing Berry curvature.
- The **DNL is topological via a loop Berry phase**, not a Chern number.
- $\mathcal{H}_{\mathcal{D}_C}$ tunes between a nodal-line magnon phase and a Magnon chern insulator phase.

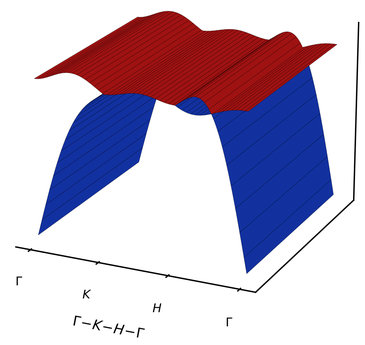

Figure 5: Visualization of Topological DNL Phase due to $\mathcal{PT}$-Symmetry in absence of $\mathcal{D}_C$

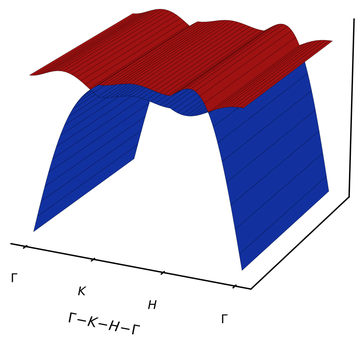

Figure 6: Visualization of Gapped Magnon Chern Insulator Phase due to $\mathcal{PT}$-Symmetry Breaking in presence of $\mathcal{D}_C$

Future Directions:

We will identify and Connect to **concrete material candidates** and experimental signatures in thermal transport and INS. We further aim to explore such systems with vanishing Berry Curvature and try to explore **Intrinsic Non Linear Hall responses** i.e. capture **Quantum Metric Dipole contributions** to the Thermal Hall Response. This is beneficial for **magnon based spin-transport devices.**

References

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Contact Information

- Email: samriddha22@iiserb.ac.in